Homework 5 — Classical ML Baseline for Skin-Lesion Classification

Daniela Gonzalez · Computer Vision and Speech Recognition — EADA · Dataset: MILK10k (5,240 lesions, dermoscopic + clinical image each)

Repository: https://github.com/Dgbaquero123/milk10k-eda

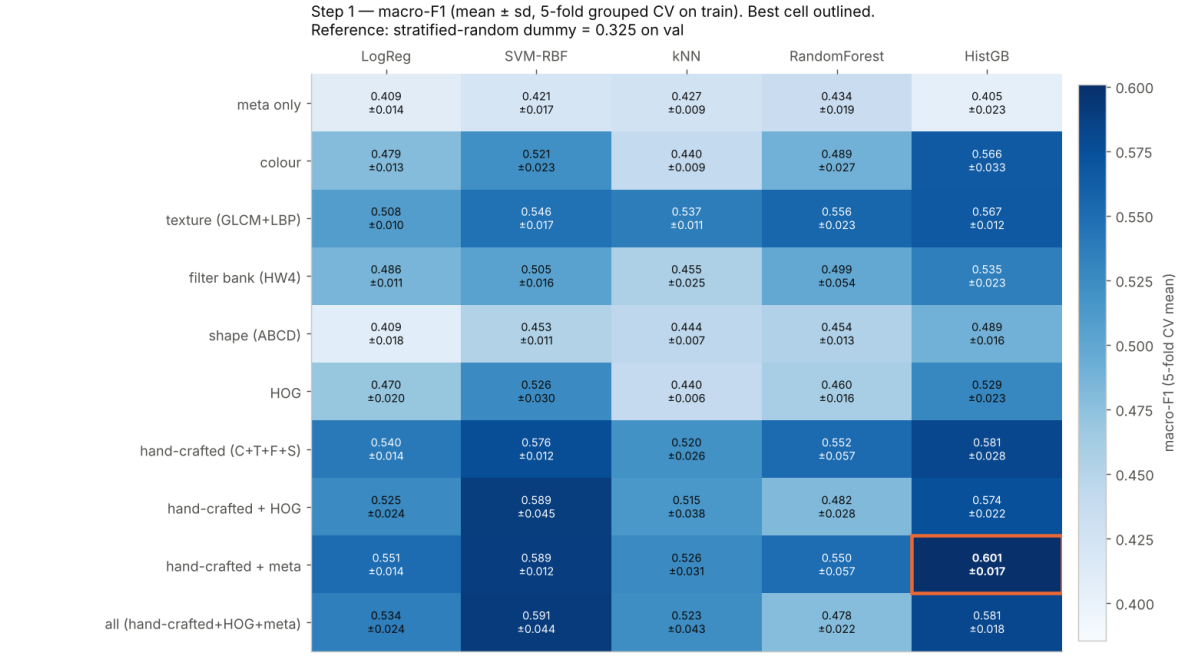
Links to the code

Deliverable	Link
Main notebook (code, results, written analysis)	github.com/Dgbaquero123/milk10k-eda/blob/main/notebooks/homework_part_5.ipynb
Notebook exported to HTML	github.com/Dgbaquero123/milk10k-eda/blob/main/notebooks/homework_part_5.html
Feature extraction module (colour, texture, HW4 filters, shape, HOG)	github.com/Dgbaquero123/milk10k-eda/blob/main/milk10k_pipeline/features.py
Baseline module (feature sets, models, CV, metrics, CIs, cost rule)	github.com/Dgbaquero123/milk10k-eda/blob/main/milk10k_pipeline/baseline.py
Feature extraction script	github.com/Dgbaquero123/milk10k-eda/blob/main/scripts/extract_features.py
Unit tests for features and decision rules	github.com/Dgbaquero123/milk10k-eda/blob/main/tests/test_features.py
Figures	github.com/Dgbaquero123/milk10k-eda/tree/main/reports/figures_hw5
Result tables (CSV)	github.com/Dgbaquero123/milk10k-eda/tree/main/reports/hw5_results
Splits reused from Milestone 1 (lesion-level, grouped)	github.com/Dgbaquero123/milk10k-eda/tree/main/splits
Submission index for all homeworks	github.com/Dgbaquero123/milk10k-eda/blob/main/SUBMISSION.md

What was done

Task. Predict *diagnosis_1* (Benign / Indeterminate / Malignant) per lesion. **Features** (484 per image, both views concatenated per lesion): colour statistics and histograms in RGB/HSV/Lab with lesion-vs-skin contrast; GLCM and LBP texture; the Homework 4 filter bank (Gaussian, Sobel, LoG, Gabor, Canny); ABCD-style shape of an Otsu lesion mask; HOG; plus metadata (age, sex, site). **Protocol.** Milestone 1 lesion-level splits (no sibling leak). 10 feature sets × 5 models (logistic regression, SVM-RBF, k-NN, random forest, histogram gradient boosting) compared with 5-fold StratifiedGroupKFold on train; the two best models tuned with GridSearchCV in the same CV; final choice on val; refit on train+val and test used once, with 1,000-sample bootstrap 95% CIs. All preprocessing inside sklearn Pipelines. Primary metric macro-F1; also balanced accuracy, macro AUC, per-class recall and the A3.4 clinical cost.

Model selection (5-fold CV on train)



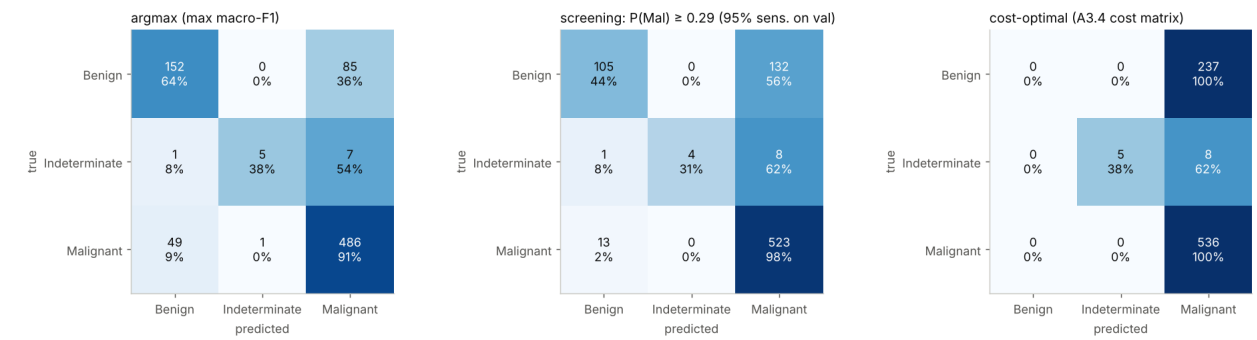
Best feature set: **hand-crafted + meta** (323 features). Tuned CV macro-F1: HistGB 0.611, SVM-RBF 0.591. Final model chosen on val: **Ensemble(HistGB+SVM-RBF)** (val macro-F1 0.563). Clinical view alone is clearly weaker (0.550) than dermoscopic alone (0.586); both views together score 0.601, a gain within CV noise.

Final results on the test split (786 lesions)

Metric	Baseline, argmax	Baseline, screening (t = 0.29)	Dummy: always Malignant	Dummy: stratified random
macro-F1	**0.697 [0.589, 0.787]**	0.644 [0.526, 0.745]	0.270	0.326
balanced accuracy	0.644 [0.556, 0.745]	0.575 [0.494, 0.678]	0.333	0.324
accuracy	0.818 [0.791, 0.844]	0.804 [0.777, 0.832]	0.682	0.561
macro AUC (one-vs-rest)	0.879 [0.844, 0.909]	0.879 [0.844, 0.909]	0.500	0.500
recall Malignant	0.907 [0.882, 0.930]	0.976 [0.962, 0.987]	1.000	0.703
recall Benign	0.641 [0.580, 0.702]	0.443 [0.379, 0.508]	0.000	0.270
recall Indeterminate	0.385 [0.125, 0.667]	0.308 [0.077, 0.600]	0.000	0.000
expected cost per lesion (A3.4)	3.282 [2.520, 4.138]	1.052 [0.655, 1.542]	0.384	9.776

Values with 95% bootstrap confidence intervals. The screening threshold on P(Malignant) was chosen on val for 95% Malignant sensitivity.

Step 5 — test confusion matrices (counts and row-normalised recall)



Conclusions

• The baseline clearly beats all dummies: test macro-F1 0.697 [0.589, 0.787] and macro AUC 0.879 vs 0.33 and 0.50.
• Colour and texture of both views are the most useful cues; shape and metadata add a little; HOG adds noise; the dermoscopic filter bank is redundant once texture is present.
• Gradient boosting and SVM-RBF are the strongest classical models; their soft-voting ensemble generalises best on val.
• As a triage tool at 97.6% Malignant sensitivity it still spares 44% of benign lesions, but it is not confident enough to rule out malignancy under the 50:1 cost matrix.
• Uncertainty is large (13 Indeterminate test lesions; a tiny feature perturbation moved test macro-F1 from 0.697 to 0.722), so differences below about 0.02 are noise. The CV estimate (about 0.61) is the most reliable single number. This is the reference a CNN must beat in Milestone 2.